

Student Guide

Term 3

4.5 ECTS

Image Matrix Transformer: A Linear Algebra Approach to Image Manipulation

Twente Pathway College

2025-2026

Contents

1.	Educational Programme/stream	3
2.	Project team	3
3.	Content description of the project	3
	<i>Task 3: Image Transformation Functions</i>	5
	<i>Task 4: Integration and Main Program</i>	5
4.	Requirements	8
5.	Assignments	8
6.	Programme outcomes and specific project objectives	11
7.	Teaching methods within the project and timeline	13
8.	Assessments within the project	15
9.	Project weighting system	15
10.	Penalties & Resits	16
11.	Query procedure:	17
12.	Use of AI	17

1. Educational Programme/stream

Engineering and Technology

2. Project team

Imre van Veldhoven	<i>Computer Science Teacher</i>
Olga Karageorgiou	<i>Soft Skills Teacher</i>
Nancy	<i>Math Teacher</i>
Christy Albaugh	<i>ALS Teacher</i>
Daniel Gomez	<i>Project tutor</i>
Thari Zweers	<i>Project tutor</i>

3. Content description of the project

Introduction

Image Processing is an interesting field. With lots of image processing software available in the market, you can easily make changes to any image you want. But how does this image processing software work, and how can we design our own software to do image processing? It looks like a big thing at this stage to design our own software. This is possible with the knowledge that you are going to gain in Term 3. The matrices needed for this project are used in almost all branches of engineering. For example, in aerospace engineering, the maximum load capacity of helicopter rotor blades is found by using matrices, in civil engineering matrices are used to understand the stresses and strains, in computer science they are used in image processing. Studying matrices without any practical relevance can be boring sometimes but through this project you will find it more and more interesting. Loops are used mainly to perform repetitive tasks. Through this project, you are expected to be an expert in the usage of loops. You will use Soft Skills to reflect on your own improvement within the project and present all your work as per ALS standards.

Project work is a valuable approach to learning that plays a pivotal role at the University of Twente. Emphasizing a practical approach, project work equips you with essential skills and experiences. Its usefulness lies in fostering interdisciplinary learning, promoting teamwork, and creating a sense of responsibility in forging students' learning paths combined with Soft Skills. While working on project assignments, you will learn to work in teams, deal with conflicts, and develop your problem-solving skills as you work collectively to overcome these challenges, all of which will help you at the University of Twente and later in your professional career.

Task

In this project, students can work in **a team of 6 students** to design and write a program that can transform an image. Students have the freedom to choose the team they want to work with within the first Project lecture of Term 3.

Technical Description: The project can be split up into two parts:

1. Part X: Linear Algebra Toolkit

Description: Students must implement fundamental matrix operations in Python using only basic programming constructs (loops, conditionals, functions, and lists). Note: Students are not allowed to use any built-in matrix manipulation functions or libraries for these operations; you must write your own functions from scratch.

- Reinforce the use of basic control structures (loops, conditionals) and functions.
- Practice working with Python data types (lists, integers, floats) in the context of linear algebra.

- Develop your own implementations of the matrix operations: addition, subtraction, multiplication, transposition, and scalar multiplication.

Plan: This work is divided into five tasks. For each task, you are required to:

- Write the function using only the basic programming constructs.
- Make error handling (e.g. not correct dimensions of matrix of vectors).
- Test your function with sample input matrices.
- Document your code by commenting explaining your logic.

Tasks and Function Signatures:

Task 1: Matrix Addition

Objective: Write a function that adds two matrices of the same dimensions elementwise.

Task 2: Matrix Subtraction

Objective: Write a function to subtract matrix B from matrix A elementwise.

Task 3: Matrix Multiplication

Objective: Write a function that multiplies two matrices.

Task 4: Matrix Transpose

Objective: Write a function to transpose a matrix.

Task 5: Scalar Multiplication

Objective: Write a function to multiply every element of a matrix by a scalar value.

2. Part Y: Image Transformation

Description: This will be an extension of your linear algebra toolkit module. You will apply your matrix manipulation functions to perform image transformations. The key idea is to convert a bitmap image into a matrix (2D list of pixel values), apply a transformation using your own code, and then convert the matrix back into an image. The different image processing functions you should build are:

rotation around a certain point; scaling; skewing; transposition; changing colour to greyscale and edge detection.

Important: You are not allowed to use Pillow's built-in image transformation methods (e.g., rotate, transpose, resize, etc.). Only *Image.open* and *Image.new* may be used for image file handling.

Plan: The assignment consists of the following steps:

- Image to Matrix Conversion:** Write a function to convert an image to a matrix. Use a `PIL .Image` method to convert a bitmap image into a 2D list of pixel values. That should be a python 2D list (Matrix)
- Matrix Transformation:** Implement function signatures for the transformation operations rotation, scaling, skewing, and transposition using your own matrix toolkit and the operations colour to greyscale and edge detection. *Do not use any Pillow transformation functions.*
- Matrix to Image Conversion:** Write a function to convert the matrix back to an image. Use a `PIL .Image` method to convert the transformed matrix back into an image.
- Integration:** Create a main program that loads an image, converts it to a matrix, applies a chosen transformation, and saves the resulting image.

Tasks and Function Signatures:

Task 1: Image to Matrix Conversion

Objective: Convert a colour bitmap image into three matrices (red, green and blue) of pixel values.

Task 2: Matrix to Image Conversion

Objective: Convert the RGB matrices of pixel values back into an image that can be shown and saved by python image functions.

Task 3: Image Transformation Functions

Objective: Implement functions for performing image transformations on the matrix. These functions should use your own matrix operations and should not rely on Pillow's built-in transformation methods.

Transformations:

- Rotation
- Scaling
- Skewing
- Transposition
- Colour to greyscale

For the more complex transformation of edge detection, you need to write a python function, this code doesn't need to use only your own defined matrix transformations. It is still not allowed to rely on Pillow's built-in methods. Pay attention to the fact that this function is more complex than the others, start on time.

Task 4: Integration and Main Program

Objective: Write a main program that:

- a. Loads a bitmap image using Image.open.
- b. Converts the image to a matrix using your image to matrix function.
- c. Applies one of the transformation functions (e.g., rotate matrix).
- d. Converts the transformed matrix back to an image using matrix to image.
- e. Saves the resulting image using Image.new and the save method.

Starter Code:

```
def main():
    # TODO: Load the image using Image.open.
    # TODO: Convert the image to a matrix.
    # TODO: Ask the user for the type of transformation and corresponding parameters.
    # TODO: Call the appropriate transformation function.
    # TODO: Convert the matrix back to an image.
    # TODO: Save the output image.
    pass

if __name__ == "__main__":
    main()
```

Starter code for all functions:

```
def matrix_addition(A, B):
    """
    Add two matrices A and B and return a
    new matrix that is the result of A + B.
    If A and B are not correct dimensions
    give an error.
    """
    pass

def matrix_subtraction(A, B):
    """
    Subtract matrix B from matrix A and return
    a new matrix that is the result of A - B.
    If A and B are not correct dimensions give
    an error.
    """
    pass

def matrix_multiplication(A, B):
    """
    Multiply two matrices A and B and return a
    new matrix that is the result of A x B.
    If A and B are not correct dimensions
    give an error.
    """
    pass

def matrix_transpose(A):
    """
    Transpose the matrix A. Returns the transposed
    matrix of A
    """
    pass

def scalar_multiplication(A, k):
    """
    Multiply every element of the matrix A by a
    scalar value k. Returns a new matrix that is
    the result of k x A.
    """
    pass

def image_to_matrix(image):
    """
    Convert a color image to three matrices of pixel values.
    The matrices have the pixel values of either red, green or
    blue. Input is an image object loaded by using Image.open.
    """
    pass

def matrix_to_image(matrix_red, matrix_green, matrix_blue):
    """
    Convert the three matrices of pixel values back to an
    image object. Output is an image object that can be
    used with img.show() function.
    """
    pass
```

```

def rotate_matrix(matrix, point, angle_degrees):
    """
    Rotate the matrix by a given angle (in degrees) around
    the specified point. The point is a vector with the
    x and y value. Do not use Image.rotate or similar methods.
    Output is a new matrix after rotation, this matrix can
    have different sizes.
    """
    pass

def scale_matrix(matrix, scale_factor):
    """
    Scale the matrix up or down by the given scale factor.
    Do not use Image.resize or similar built-in methods.
    Output is a new matrix after scaling.
    """
    pass

def skew_matrix(matrix, skew_x, skew_y):
    """
    Skew the matrix by given factors in x and y directions.
    Do not use any built-in transformations.
    Output is a new matrix after skewing.
    """
    pass

def transpose_matrix(matrix):
    """
    Transpose the matrix (swap rows and columns).
    Do not use any built-in transformations.
    Output is a new matrix after skewing.
    """
    pass

def to_grayscale(matrix_red, matrix_green, matrix_blue):
    """
    Change the color image to a grayscale image.
    Do not use any built-in image methods.
    Output is a single matrix with values between 0-256
    in such a way that it can be used with the img.show()
    function.
    """
    pass

def edge_detection(matrix):
    """
    Find the edges in a grayscale image.
    Do not use any built-in image methods.
    Output is a single matrix where all values are 0
    except for the edges, those values have 256.
    Make sure that the output is such that it can be
    used with the img.show() function.
    Tip: This function is way more complex than the
    other functions, make sure to start on time with
    this!
    """
    pass

```

4. Requirements

For designing an image processing software, you are required to have knowledge of:

- Matrices and their transformations: You will learn about the “Matrices” and various transformations in the mathematics course in term 3.
- Loops, control structures, and basic Python algorithms: You will learn about these topics in your Computer Science course in term 3.

Scope: Image processing is also an art where you can add your creativity to the project with some simple matrix operations. Thus, you are allowed to add as many features as you want to your software. The minimum features that your project should have are all the functions defined above.

Tip: Images must be converted to bitmaps, as modern formats like JPEG or PNG use algorithms beyond the level of this course.

Important Note:

1. We are aware that this project can be made very easy with the use of various commercially available AI tools, but then the project will lose its significance, and so will your learning objectives. Thus, you are required to build everything on your own. If any use of AI is detected, the entire group will be penalised.
2. As all of you are going to learn matrices and Loops, all of the group members are expected to be aware of the code and should be able to explain if requested during any process of the evaluation.

5. Assignments

For all the assignments in this project, please consult the relevant marking rubric on Moodle to see what you will be graded on. This project contains the following assignments:

Week 2: Team Charter Paper (formative):

If you would like to receive feedback on your team charter paper, you can submit it on Moodle by the 2nd of April at 20:00. This is an optional assignment, but it may be helpful to receive some feedback before you start the project.

Week 5: Panel discussion:

With the whole project group, you will discuss your linear algebra toolkit (part X). Both Computer Science and Mathematics will be assessed during this discussion. You must hand in the code the day before the presentation on Moodle, and you should be able to explain it and answer questions about the code and the mathematics of the matrix operations behind it. Every group member should be able to answer all questions. The discussion is with the whole group, and students will be assessed **individually** according to their individual knowledge of the project. As well as how well the code works.

This panel discussion should last a maximum of 20 minutes. All group members must be present and answer the questions. You do not need to prepare any slides for this part of the assignment.

Soft Skills: For the Soft skills, you will have to evaluate your time management in your group so far, as well as how you plan to ensure good cooperation and task focus throughout the rest of the term. What will you do from now on to ensure an owl approach (results & relationships)? The Soft skills part must be included in a pdf. file you submit on Moodle, but **it will not be evaluated for its APA style**. There is no word count limit. You will be assessed as a **group** for this criterion. See the rubric for more information.

Week 5: Peer evaluation part 1

As part of this group project, you are required to complete a peer evaluation to assess both your own contributions and those of your team members. The evaluation will be individually submitted

by each student. The ratings you give will be averaged across the group members and will count for 5% of your total project grade. You are expected to provide honest and objective assessments based solely on each member's project performance. This ensures fairness and allows the teachers to recognize individual contributions more accurately. The Form will be sent to you by email. Use the form by following the instructions carefully.

If the average of the 2 Peer evaluations is below 55%, you may have to resit the project; this is dependent on your peers' assessment of your performance.

Week 9: Python Documentation

Once you have written and tested your code, you must, as a group, write a documentation for this library (both X and Y), and some general background. This documentation helps other users to utilise and understand your code. This documentation should at least include (but is not limited to):

- General background for Python/programming languages and their use. Must cite 3 academically appropriate sources.
- Explanation of the purpose of your code (what it does)
- Clear and detailed explanation of how the code works.
- Examples of how to use your code

ALS:

Your documentation should be 2000 words (+/- 10%), word count must be on the title page and not including References and Appendix (required sections); and include at least 4 academically appropriate sources in a general background section. You may, of course, include other sources as well, but a minimum of 4 CRAAP-approved sources of information.

You must decide what sections to include, and how long they each are, but part of what is graded is how well-organised your documentation is. This is not strictly a report, but you may structure it in a similar fashion.

You can use your creativity for the design of your documentation (colours, fonts- stick to readable fonts, margins, spacing, pictures) but **please make sure you follow APA standards for citations, references, images and figures**. You are required to submit your Python code (in working condition) along with the project report, as an Appendix.

Math: In addition to explaining the programming logic, students must clearly explain the mathematical operations and transformations implemented in their code.

The documentation must include:

- The matrix representations used for image transformations such as rotation, scaling, skewing, and transposition.
- An explanation of how pixel coordinates are represented mathematically (e.g. as coordinate pairs or vectors).
- A step-by-step explanation of how matrix operations (addition, multiplication, scalar multiplication, transposition) produce the final transformed image.
- A clear connection between the mathematical expressions and their Python implementation (e.g. how loops correspond to matrix multiplication).
- For colour transformations (e.g. greyscale), a brief explanation of the mathematical method used to combine RGB values.

Simply describing what the code does is not sufficient; students must explain how the code works mathematically.

You must submit the documentation as a **.pdf file** before the deadline on Moodle.

Word count deductions:

- 5% deduction to the assignment grade if it is between 151 to 250 words under/over the word limit

- 10% deduction to the assignment grade if it is between 251 to 350 words under/over the word limit
- 15% deduction to the assignment grade if it is above 351 words under/over the word limit

Week 10: Peer evaluation part 2

As part of this group project, you are required to complete a peer evaluation to assess both your own contributions and those of your team members. The evaluation will be individually submitted by each student. The ratings you give will be averaged across the group members and will count for 5% of your total project grade. You are expected to provide honest and objective assessments based solely on each member's project performance. This ensures fairness and allows the teachers to recognize individual contributions more accurately. The Form will be sent to you by email. Use the form by following the instructions carefully.

If the average of the 2 Peer evaluations is below 55%, you may have to resit the project; this is dependent on your peers' assessment of your performance.

Week 11: Final Presentation:

You will present your results and any obstacles. A clear explanation of the research background, methodology, and analysis of the program must be provided. The Q&A performance will be considered in the grade. You will be assessed on Computer science theories, the functionality of the program, and algorithm analysis, as well as ALS presentation skills, APA rules (make sure you label figures and tables!) and Mathematics knowledge.

Your presentation should last a maximum of 10 minutes, followed by 10 minutes of Q&A. You can choose 3 students to present. All group members must be present and answer the questions.

You must upload the slides to Moodle the day before your presentation at 20:00.

Word count deductions:

- 5% deduction to the assignment grade if it is 3 minutes under/over time
- 10% deduction to the assignment grade if it is 4 minutes under/over time
- 15% deduction to the assignment grade if it is 5 minutes under/over time

Week 12: Resit (if needed):

Panel discussion: You and/or your group will attend a panel discussion with the teachers.

Create a 10-minute presentation (PowerPoint/Canva) about your project. Both individual and group resits require this presentation. Note that this must not be a new project you created by yourself- it should be an *improved* version of what you and your group worked on throughout the term. You will need to:

- **Maths:** Students must explain the mathematics behind their code, including matrix representations for image transformations, mathematical representation of pixel coordinates, and how matrix operations produce the final image. Clear links between mathematical expressions and their Python implementation are required. During the Q&A session, answers must be clear and mathematically justified.
- **CS:** Show a deep understanding of the code. During the resit, the focus will be on part Y, but part X is also assumed to be known (and can be questioned). You should be able to explain all the different aspects of the code. How it works, how it is used and why you made certain choices.
- **ALS:** Display appropriate presentation skills. Answer questions regarding APA formatting, report sections, and the purpose/use of academic sources in research. Be able to explain tables and figures presented in the documentation.
- **Soft Skills:** Reflection on your team's collaboration framework, your personal contribution, an evaluation of all team collaboration aspects that worked or did not, as well as possible adjustments for the next project work.

The Q&A will last 20 minutes maximum. You should expect at least one question per project objective stated below. Should the entire group be present, all members should have equal speaking time during the presentation and Q&A session. You must submit your slides on Moodle the day before the Resit.

The presentation should last 10 minutes, and the following deductions apply for going under/over time:

- 5% deduction to the assignment grade if it is 3 minutes under/over time
- 10% deduction to the assignment grade if it is 4 minutes under/over time
- 15% deduction to the assignment grade if it is 5 minutes under/over time

This resit assignment (only allowed if you scored under 55%) is not mandatory (unless you failed the Peer Evaluation and received a poor rating from your team members), however, if two or more members of your team also want to resit the project, you must do it together.

6. Programme outcomes and specific project objectives

In this project, the following programme outcomes and specific project objectives will be addressed and assessed. Aligned with that the specific project objectives are shown in relation to the level of difficulty according to Bloom's taxonomy:

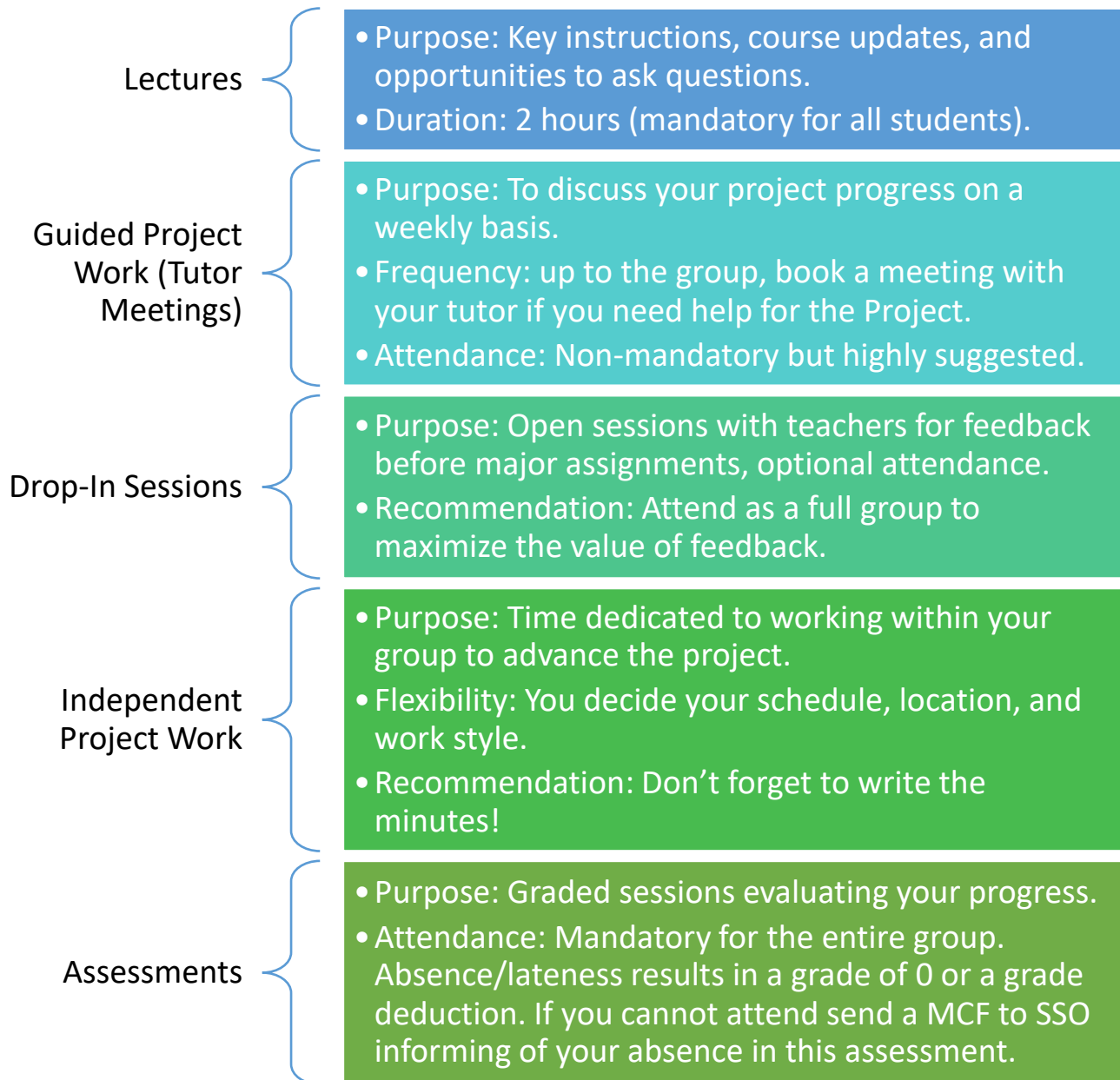
Programme outcomes	Specific project objective After completing this project, students will be able to:	Level of difficulty according to Bloom's Taxonomy*
Write clearly, coherently and logically in a variety of formats, while adhering to standard academic conventions.	1. Write a clear, grammatically accurate, lexically diverse, and detailed instruction manual in an appropriate academic format following the APA referencing format.	Application
Locate, read, summarize, evaluate and reflect on academic texts	2. Present information and communicate orally with visual aids in an academic context concerning your project following the APA referencing format.	Application
Communicate confidently and clearly in a variety of formal and semi-formal oral formats		
Process, evaluate, exploit and organize information provided to them in spoken English		
Write clearly, coherently and logically in a variety of formats, while adhering to standard academic conventions	3. Develop a realistic timeline and contingency plan to manage tasks in groups.	Application

Communicate confidently and clearly in a variety of formal and semi-formal oral formats		
Operate effectively in a range of social and cultural contexts both in class and outside	4. Critically evaluate self and peers by assessing contributions to the project.	Evaluation
Demonstrate application of mathematical techniques in the fields of combinatorics and probability, complex numbers, and linear algebra, at a beginner undergraduate level.	5. Matrix operations (addition, subtraction, multiplication, transpose, scalar multiplication) and transformations.	Application
Obtain a general understanding of computing, software and programming languages, algorithms, pseudocode, and flowcharts.	6. Develop a Python program that incorporates user input and output operations effectively, employing control structures such as conditions and loops to create a functional and efficient application.	Application
Translate programming problems into code using basic programming skills.	7. Apply programming concepts and skills, such as defining variables, using control structures, creating functions, and employing data structures, to successfully transform various programming challenges and tasks into well-organized, executable code solutions for real-world situations.	Application

* knowledge, understanding, application, analysis, synthesis, evaluation

7. Teaching methods within the project and timeline

Throughout this course, you'll engage in several activities to support your learning. While guidance is provided at key moments, the majority of the project work is independent and done within your group. Here's an overview of the course structure:

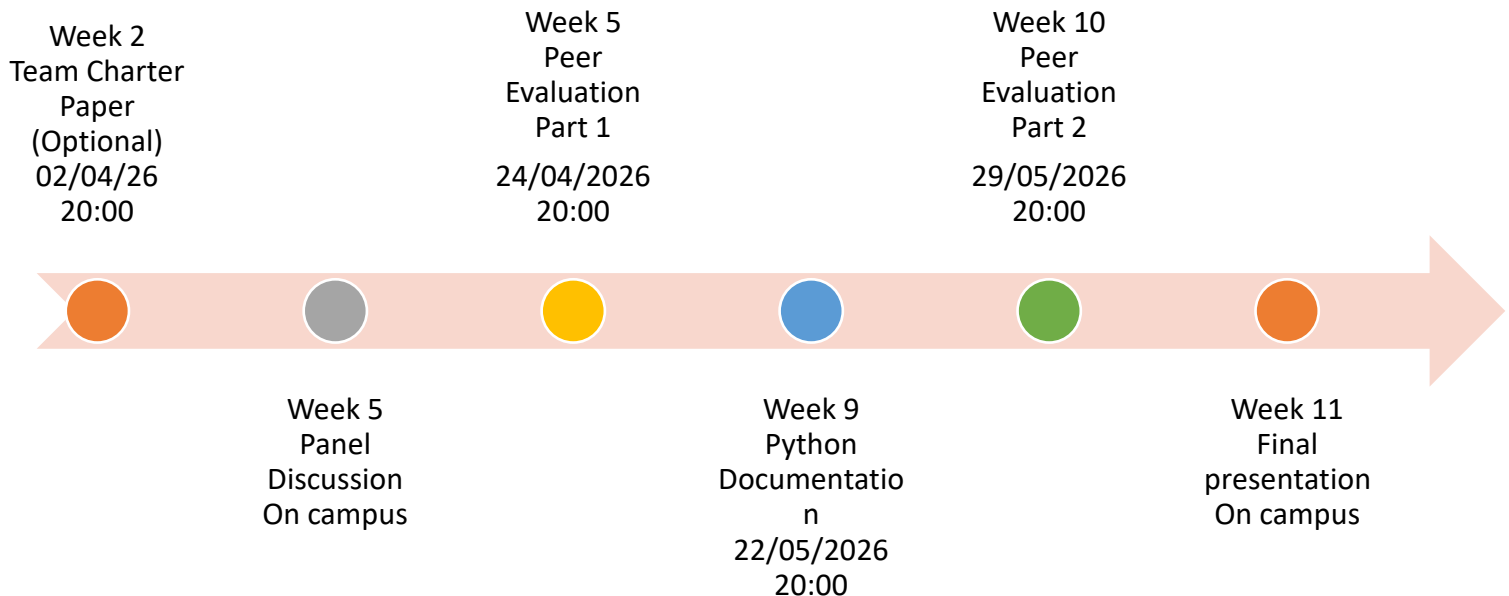


The table below gives an overview of the teaching methods and how many hours the different teaching methods occupy per week, within the last column the total number of hours per teaching method (and percentage of total). These numbers are an approximation.

Teaching methods in hours per week	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	Total
Lectures	2	0	0	0	0	0	0	0	0	0	0	0	2
Workshops	0	0	0	0	0	0	2	0	0	0	0	0	2
Guided project work	0	0	0.5	0.5	0	0	0	0.5	0.5	0	0	0	2
Drop in session	0	0	0	0	0	0	0	1	0	0	0	0	1
Project work	10	10	12	12	12	10	12	12	12	7	6	0	115
Assessment	0	0	0	0	1	0	0	0	0.5	0.5	2	0	4
Total number of hours	12	10	12.5	12.5	13	10	14	13.5	13	7.5	8	0	126

Timeline

The timetable of the project can be found on Moodle. Changes may occur, so check the timetable regularly.



8. Assessments within the project

Assignments	Group/individual	Weight	Specific Project objective (see section 6)
Team Charter Paper (formative)	Group assignment/Optional	-	3, 4
Panel discussion	Individual and Group assignment	15%	3, 5, 6, 7
Python Documentation	Group assignment	65%	1, 5, 6, 7
Peer evaluation*	Individual assignment	10% Part 1 – 5% Part 2 – 5%	4
Final Presentation	Group assignment	10%	2, 5, 6, 7

*If the average of your two Peer Evaluation scores is below 55%, you may fail the project and receive a 0% grade. This outcome depends on the scores your peers have given you. If you do receive 0% due to failing Peer Evaluation, you must resit the project to pass the term.

Students who submit the peer evaluation after the deadline, or fail to submit it, will retain the peer-assigned grade (rather than receiving a zero). However, a standard 10% deduction will be applied in accordance with the late or missing assignment policy.

Important: If you receive a good overall score from your peers but fail Peer Evaluation because you forgot to submit it, you will not automatically receive 0% for the project. In this case, you will keep your final project grade (including the usual penalties for not submitting an assignment).

Examples:

Panel discussion	Python Documentation	Peer Evaluation (avg of Part 1 and 2)	Final Presentation	Final term grade	Outcome
75.00%	59.00%	80.00%	60.00%	63.60%	Pass
65.00%	70.00%	50.00%	89.00%	69.15% = 0%**	Fail – must resit
48.05%	55.55%	40.95%	40.15%	51.43%	Fail – should resit

**If you received a low grade from your peers.

9. Project weighting system

Students will receive one overall Project grade (%) in June/July. This grade will be an average of the student's project grades over the three terms. **In order to pass TPC, you must pass the Project course.**

	Weight (%)
Term 1	30
Term 2	30
Term 3	40

To pass the Project course, the overall Project grade must be 55% or above.

See, for example, two possibilities:

Term 1	Term 2	Term 3	Average	Outcome
29.00%	80.00%	71.00%	60.00%	Pass
50.00%	52.00%	60.00%	54.00%	Fail
55.00%	55.88%	54.02%	54.87%	Fail

10. Penalties & Resits

Penalty policy:

Reasons to receive a penalty:

- Failure to submit any assignment by the initial deadline
- Failure to attend (in person or on time) any of the on-campus assignments (the presentations)
- Penalties for not submitting a group assignment on time lead to penalties for the whole team. Lack of attendance to an individual assignment leads to an individual penalty.

Any of these actions will result in a 0 for that assignment and a penalty on the final grade. Thus, each missing/late assignment (or missing tutor meetings) would have a 0 plus an additional 10% points off the term project grade. See the table below:

Penalties	Total deduction*
1	10% points
2	20% points
3	30% points
4	40% points
...	...

*Note: The minimum grade is 0%.

For example,

Final project grade	Failed to attend Panel Discussion	Failed to submit Peer evaluation Part 1	Failed to submit report	Total deduction	Total project grade
78.56	-10	-10	-10	30	48.56

Resit:

You can only attend the resit if your term final grade is below 55.

Panel discussion: See the assignment description in the [Assignments](#) section. This resit assignment is not mandatory, but if two or more members of your team also want to resit the project, you must do it as a whole group. The maximum grade you can receive in this resit is 55.00%. If you receive a lower grade than the original, you will keep the highest grade.



There will be no official All-term resit for this course. All resits for this course occur during the Academic year.

11. Query procedure:

You may query any grades only ONCE per term, at the end of the term, **using the query form on Moodle** (mandatory).

Requirements:

- Queries should be formulated politely and respectfully.
- Queries should specify the assignment/s you wish to challenge your grade for.
- Queries must have the entire group's approval
- You should clearly explain, **using content-related arguments and the rubrics**, why your original work/answer is worth more points.

Please note that if you do not meet the above requirements, you will not receive a response.

The teachers will respond within 3 working days.

12. Use of AI

In line with UT policies and procedures, if Artificial Intelligence (AI) tools are allowed for completing an assignment several different levels are possible, i.e. only for certain purposes, only specific tools, or completely free. It is required that students mention the use of AI. Therefore, when any form of AI is used, the use of these tools should be included in the appendix (list all tools that were used during the work). The following sentence should be used at the start of the appendix:

“During the preparation of this work, the author(s) used [NAME TOOL / SERVICE] in order to [REASON]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the work.”

If there is nothing to disclose, a statement must still be included that no AI was used:

“During the preparation of this work, the author(s) used no artificial intelligence tools.”

Generally, all software that has been used needs to be disclosed (because it has the potential to include AI), for example, but not limited to, tools for writing, searching, citing, and analyzing data. Please be aware that the following programs could contain tools based on AI, generally included to improve language: e.g. (but not limited to) text processors (such as Word), spelling and grammar checks, and potentially reference managers. Students using AI, without the explicit consent of the instructor and acknowledgment of the tool in an appendix, should therefore be considered to have committed academic misconduct. Please read the academic handbook on how to understand how TPC handles cases of misconduct/academic impropriety.

Assignment	Use of AI	Description
Team Charter Project development Presentation Project report Peer and self-evaluation Symposium	Some	AI can be used for the following purposes: to check your grammar and generate references. The following AI tools can be used: • Grammarly: to check your grammar, vocabulary, and sentence structure. • Scribbr: to assist you with referencing